PACE: Peer-Wise Calibrated Experts for IoT Intrusion Detection with Benign-Only Target Data

Salaheddine Naslouby[1,*], Mohamed Lachgar[2], Hamid Hrimech[1]

[1]*Hassan First University, National School of Applied Sciences (ENSA), LAMSAD Laboratory, Berrechid, Morocco*

[2]*Cadi Ayyad University, Higher Normal School (ENS), L2IS Laboratory, Marrakech, Morocco*

Abstract

Detecting intrusions at a newly monitored IoT target is difficult when only benign target observations are available but peer clients possess labeled attacks. Combining peer scores under a single calibrated threshold can obscure useful evidence from individual classifiers. PACE retains separately calibrated peer evidence and combines the resulting alert union with a protected local anchor and a heterogeneous bank of anomaly experts. The local branch incorporates feature-rarity and peer-response evidence without using target attack labels. We evaluate PACE on 58 targets spanning six corpora and seven study strata, using three learning seeds and local fitting budgets of 100 and 400 rows. Relative to an identical peer branch paired with a seven-detector mean local ensemble, PACE improves target-macro detection by 2.61 and 2.83 percentage points at nominal $\alpha = 0.01$, and by 2.34 and 2.32 points at $\alpha = 0.05$. At the latter operating parameter, PACE also reduces mean realized false-alert rates at both fitting budgets. Ablations identify rarity-based evidence as the largest measured complementary contribution among the tested expert families. Gains are strongest in TON_IoT, with smaller improvements elsewhere. False-alert rates remain heterogeneous across targets. With the complete local branch held fixed, retaining separately thresholded peers improves detection by 3.0–7.3 percentage points over mean or maximum peer-score fusion at the tested nominal settings, with higher mean false-alert rates. PACE's gains therefore reflect complementary peer and local detection rather than a guarantee of future false-alert control.

Keywords: IoT intrusion detection, peer models, benign-only learning, empirical calibration, anomaly ensembles, false alerts

Draft for author review. Declaration confirmations and final human approval are pending.

1. Introduction

Monitoring a newly introduced target presents an asymmetric learning problem. The target can supply examples of legitimate operation, while other clients may already have

*Corresponding author: Salaheddine Naslouby. ORCID: 0009-0002-7295-7816.

Email addresses: naslouby.salahe@gmail.com (Salaheddine Naslouby), m.lachgar@uca.ac.ma (Mohamed Lachgar), hamid.hrimech@uhp.ac.ma (Hamid Hrimech)

labeled malicious traffic. A supervised detector trained only for the target is unavailable, but discarding peer attack information would leave the monitor dependent on deviations from its own benign sample. The practical question is how to preserve useful peer evidence while retaining a local route for attacks that those peers miss.

A common fusion strategy takes the largest peer score and calibrates one threshold on target benign observations. This strategy can lose discrimination when the benign tail of one peer dominates the fused score: the common threshold must accommodate that tail, and another peer's informative but numerically smaller attack evidence may no longer trigger. We call this task-specific failure mode peer drowning. Score normalization and domination by a single extreme detector in maximum fusion are established concerns in outlier ensembles [1]. Target-relative score transformations address differences in score scale, but they do not make the tails of heterogeneous experts interchangeable or ensure that a single threshold on their maximum preserves every useful detector.

PACE (Peer-wise Calibrated Experts) converts each peer's attack-minus-benign margin into target-relative upper-tail evidence, using empirical ranks on a benign reference set and extrapolation beyond its maximum. Separate empirical thresholds are calibrated on a later benign role, and a shared multiplier adjusts the peer union according to its observed blockwise alert rate.

PACE addresses this setting through two complementary design choices. First, it retains the foreign supervised classifiers as separately calibrated experts rather than collapsing their scores into a single statistic before thresholding. Second, it replaces a single mean-ensemble local decision with a protected anchor and a heterogeneous bank designed to detect attacks outside the peer union. Related calibrated hybrid detectors, particularly MCDE [2], provide important precedent; the distinction examined here concerns peer multiplicity, the target information regime, and the structure of the final decision.

The distinguishing feature is not the individual anomaly estimators or empirical thresholding in isolation. It is the use of multiple frozen supervised peer classifiers with separate target-relative thresholds, coupled with an independently calibrated local anchor and heterogeneous expert bank. This structure preserves distinct peer decisions and provides a complementary local detection route when the supervised peer union misses an attack.

The article makes three contributions:

1. A peer-wise calibrated intrusion detector that retains separately thresholded foreign supervised classifiers and combines their union with a protected local anchor and a heterogeneous anomaly-expert bank. The implemented decision rule, reference transformation, blockwise calibration, and allocation procedure are specified precisely.
2. An architectural evaluation that separates two questions: whether the complete heterogeneous local branch improves detection over a shared-peer mean-ensemble comparator, and whether retaining separately calibrated peer decisions improves detection over mean and maximum peer fusion when the local branch is fixed. The experiments report detection and realized false-alert tradeoffs across 58 targets from six corpora.
3. An analysis of complementary local detections, peer-fusion alternatives, expert-family ablations, corpus-level variation, and fixed-cohort fitting-budget changes, identifying where the method adds coverage and where its benefit is limited.

Across the 58-target primary cohort, PACE improves target-macro detection over the

shared-peer mean-ensemble comparator by 2.61–2.83 percentage points at nominal $\alpha = 0.01$ and 2.32–2.34 points at $\alpha = 0.05$. At $\alpha = 0.05$, both fitting budgets also show lower mean realized false-alert rates. The largest gains occur in TON_IoT, while the ablations identify rarity-based local evidence as an important source of complementary detections. The evaluation also characterizes target-level variability and calibration limitations rather than assuming that a nominal operating parameter guarantees a corresponding future alert rate.

2. Related work

2.1. Calibrated supervised and benign-deviation evidence

A close comparison is the Mode-Calibrated Dual-Evidence Detector (MCDE) of Yue et al. [2]. It combines a supervised histogram-gradient-boosting classifier with distances to learned benign modes, converts both scores to empirical benign-tail evidence, allocates branch shares, and calibrates a final fused threshold on a separate benign set. Its principal evaluation withholds entire attack families. A supplementary device-transfer experiment recalibrates the final threshold with trusted target benign data and exposes a false-alert versus recall tradeoff.

MCDE establishes precedent for calibrated supervised and benign-deviation fusion. PACE retains multiple foreign peer classifiers as separately thresholded experts and adds a heterogeneous local anchor and bank. It calibrates the peer union, anchor tier, and bank union separately, without a final aggregate threshold, and its low-alpha resolution rule does not preserve a strict partition of the nominal allocation. Its target-label restriction also differs from family hold-out: peers may have seen the target's attack families. The resulting comparison concerns the retained experts, the final decision structure, and the target information regime; MCDE was not run in the PACE campaign.

Table 1 summarizes the closest architectural and information-regime distinctions.

2.2. Empirical calibration and statistical guarantees

Rank-based score calibration has a substantial statistical foundation. Split conformal prediction uses held-out calibration scores and a finite-sample quantile correction to obtain marginal coverage under exchangeability; the probability averages over calibration and test sampling [6]. This is different from merely ensuring that a prescribed fraction of the observed calibration scores exceeds an empirical threshold. PACE uses the latter type of empirical constraint with a maximum over block-specific quantiles and data-dependent union scaling. Its threshold index is not the standard split-conformal correction, and its extrapolated evidence is not itself a conformal p-value.

Bates et al. [7] distinguish marginal conformal outlier p-values from calibration-conditional validity. Shared calibration observations induce dependence between test p-values, which matters for downstream multiple testing. Their conditional constructions require explicit statistical adjustments. Neither calling an equal-share detector union "Bonferroni" nor using a smoothed empirical tail establishes those properties for PACE. False discovery rate across tested observations also differs from a detector's benign false-alert probability; the quantities should not be interchanged. Formally valid p-value combination requires valid marginal inputs, even when a combination rule accommodates arbitrary dependence [8]. The union bound itself needs no independence assumption, but empirical threshold calibration does not establish the population tail bounds needed to apply it at a claimed test-time level.

Table 1: Methodological comparison of information requirements and decision mechanisms, not a performance ranking. None of the external systems was rerun in the PACE campaign. The two panels separate the five conceptual comparison fields for readability.

Method	Source-model supervision	Target information used
MCDE [2]	Labeled benign and known-attack training; benign-mode fitting	Benign reference and threshold sets; trusted target benign recalibration in device transfer.
Kitsune [3]	Unsupervised autoencoders; training traffic assumed benign	Initial local traffic for feature grouping and reconstruction learning.
DÏoT [4]	Benign device-type sequence learning across clients	Local device traffic and device type; threshold study uses benign and attack evaluation traffic.
OSDN [5]	Labeled source benign and attack categories	Unlabeled target traffic, potentially including unknown attacks; heterogeneous feature spaces.
PACE	Labeled benign and attack data at peers	Trusted benign fitting, reference, and calibration roles; pooled benign preprocessing.

Method	Fusion or adaptation mechanism	Calibration or final decision
MCDE	Supervised score plus distance to benign modes	Benign-tail alignment and branch allocation; separate final fused-score threshold.
Kitsune	Ensemble reconstruction errors feed an output autoencoder	Threshold on output reconstruction error; the paper discusses application-dependent threshold choices.
DÏoT	Federated aggregation of device-type sequence models	Low packet-symbol probabilities identify anomalies; their fraction in a packet window triggers an alert.
OSDN	Heterogeneous projections and open-set domain alignment	Shared classifier trained on known classes and generated unknown examples.
PACE	Retained peer experts plus local anchor and expert bank	Separate peer, anchor, and bank calibration; final alert OR without aggregate recalibration.

Barber et al. [9] extend conformal prediction with weighted quantiles and bounds on the loss of coverage under departures from exchangeability. Their results connect validity to the difference between the relevant data distributions and use a specifically constructed prediction rule. PACE does not implement that procedure. Its temporal partition and worst-block calibration do not by themselves import protection against drift. These distinctions motivate reporting realized false-alert distributions and limiting conclusions to the tested setting.

A recent CAN preprint by Xiang et al. [10] separates benign target adaptation from later benign calibration and applies empirical ranks to temporal block maxima. Its block length reflects estimated serial dependence and available rank resolution; unresolved dependence limits the validity claim. PACE instead takes the largest empirical quantile across calibration blocks and applies the resulting threshold to individual records. These operations use different decision units. The preprint reinforces that benign-only adaptation and block-based empirical calibration are existing alternatives to supervised target fitting; it does not supply a guarantee for PACE's peer unions. Per-detector threshold selection to regulate the alert rate of heterogeneous anomaly ensembles predates PACE [11]. PACE's contribution concerns the combination of frozen supervised peer reuse, target-relative thresholds, and a complementary local branch, rather than threshold allocation in isolation.

2.3. Established local anomaly models

PACE uses established anomaly-scoring mechanisms rather than a new base detector. Isolation forest scores the ease of isolating an observation in random partition trees [12]; local outlier factor compares neighborhood density with that of nearby observations [13]. These models encode different notions of deviation, but their diversity alone does not establish that a combined detector will improve detection at a fixed false-alert rate. That question requires evaluating the complete scoring and thresholding rule.

The seven-member comparator also includes reconstruction error from an autoencoder. Pang et al. [14] distinguish reconstruction-based learning from objectives designed directly for anomaly scoring: good compression of dominant regularities need not preserve the information needed to separate anomalies. This motivates treating reconstruction as one source of evidence, but it does not explain PACE's measured gains. PACE's six-member anchor omits the autoencoder while changing local calibration and adding a bank, so the primary result cannot isolate an autoencoder effect.

2.4. Benign-only monitoring, transfer, and personalization

Benign-only target modeling is established in IoT security. N-BaIoT fits device-local autoencoders to benign traffic and uses separate benign observations to choose detection settings [15]. Its nine-device Mirai and BASHLITE evaluation is relevant precedent, but its packet and window decisions cannot be ranked directly against the present target-macro evaluation. DÏoT instead learns device-type traffic models collaboratively through federated aggregation [4]. PACE assumes supervised source models trained on labeled peer attacks and retains those models separately; it does not perform iterative federated optimization.

FedMSE [16] learns benign-only autoencoder representations with centroid anomaly scoring and averages model parameters using inverse reconstruction error on a shared processed benign set. Its collaborative learning differs from retaining PACE's separately trained supervised

peers. The distinction concerns the learned representation, aggregation rule, and information access.

Rey et al. [17] evaluate federated benign-only autoencoders and supervised classifiers on N-BaIoT, including devices unseen during training. PACE instead retains distinct frozen supervised peer models and adapts their evidence and thresholds to trusted target benign data.

Recent personalization methods also learn client-specific model parameters. The federated hypernetwork IDS of Abhijit et al. [18] updates a shared hypernetwork and client representations under labeled classification objectives. PACE personalizes evidence and thresholds from target benign observations while freezing the peer classifiers. The methods therefore personalize different parts of the detector under different information regimes. The centralized pooled preprocessing used in PACE's benchmark further limits any literal model-only deployment interpretation.

Open Set Dandelion Network (OSDN) learns from labeled source data and unlabeled target data through heterogeneous projections and domain alignment [5]. Its target data can include unknown attacks. PACE instead requires trusted benign target roles and a compatible feature schema for the supplied peer models. The comparison distinguishes two attack-label-free target settings: adaptation on mixed unlabeled traffic with learned cross-schema alignment, and adaptation on trusted benign traffic with frozen compatible peer models.

DI-NIDS learns domain-invariant representations from labeled source and unlabeled target traffic before fitting a one-class SVM [19]. PACE instead calibrates compatible frozen peer models on trusted target benign data.

Calibration also has different meanings across IDS studies. Talpini et al. [20] federate parameters of sigmoid probability calibration using labeled local calibration data and a classification loss. Their goal concerns the reliability of class probabilities. PACE calibrates alert thresholds against benign empirical tails. A better-calibrated class probability need not meet a chosen benign false-alert rate, and an empirical tail threshold need not yield calibrated posterior probabilities.

2.5. Fusion and detector-response representations

Zimek et al. [1] discuss detector diversity, score normalization, and the sensitivity of maximum fusion to an individual overestimating detector. PACE's peer-wise thresholds address a particular consequence of these issues when transferred peer scores are calibrated on a new target. The shared union multiplier still couples the peers: separate thresholds do not make the final decision immune to another expert's behavior.

Modeling joint detector responses also has direct network-security precedent. Kitsune passes the normalized reconstruction errors of an autoencoder ensemble to another autoencoder, allowing relationships among first-layer errors to influence anomaly detection [3]. PACE's response expert instead fits an isolation forest to target-relative evidence from foreign supervised peers. The choice of response coordinates and estimator differs; the general idea of anomaly detection in detector-output space is not new. ViM combines classifier logits with feature-space residual information for out-of-distribution detection [21], providing a further example of complementary output and representation evidence. Its vision evaluation and geometry differ from PACE's peer-response forest and do not constitute a direct IDS baseline.

2.6. Transfer evaluation and operating points

Recent cross-protocol evidence reinforces the need to separate ranking from threshold transfer. Jin [22] evaluates IP-to-BLE adaptation using unlabeled target data, contrasting random-window evaluation with a capture-wise hold-out and auditing source-selected thresholds. Improvements depend on the split, while transferred thresholds can produce severe false-alert failures. That setting differs from PACE's trusted target-benign threshold adaptation and within-stratum feature compatibility. Nevertheless, it illustrates why a pooled detection gain or an unlabeled adaptation procedure does not establish safe operation on a new capture. PACE's false-alert distributions and preprocessing dependencies must be evaluated on their own evidence.

Ha and Kim [23] calibrate benign-validation thresholds at specified false-positive rates and transfer them unchanged to held-out attacks in host-session graph detection. They report realized false-positive rates relative to the requested level and examine failure in the tails of that distribution. Their models use labeled benign and seen-attack training examples; PACE instead studies a benign-only target's reuse of separate supervised peers and calibration of their alert union. The shared principle of assessing a fixed threshold on unseen attacks does not make the studies' aggregation units or numerical gains directly comparable.

Nouman et al. [24] trace a fixed-threshold false-positive difference to calibrating window-averaged scores and evaluating per-flow scores. PACE calibrates and evaluates row-level evidence; the external diagnosis illustrates the need to check score granularity, without identifying the cause of PACE's false alerts.

3. Problem formulation

A target is a benchmark client with benign observations and no attack labels available to its detection pipeline. Depending on the corpus, this client is a device, host, or capture scenario. We therefore use "target" rather than treating every client as an independently sampled physical device. Each target has a feature space shared with its eligible peers within its study stratum. Models are not transferred between incompatible corpus feature schemas.

Let T_N denote the first N observations of the target's retained benign training role, R its benign reference role, and C its benign calibration role. A separate benign test role and held-out attack observations are used for evaluation. Peer k supplies a supervised binary classifier with attack-minus-benign logit margin $m_k(x)$. The target's own supervised classifier, if constructed for its role as a peer to other targets, is excluded from its detector. The objective is to combine transferable attack evidence with target-local anomaly evidence without fitting or calibrating on the target's attack labels.

For an alert rule $A_t : \mathcal{X}_t \to \{0, 1\}$, the true-positive rate (TPR) and false-positive rate (FPR) of interest are

$$\mathrm{TPR}_t = \Pr\{A_t(X) = 1 \mid Y = 1, t\}, \qquad \mathrm{FPR}_t = \Pr\{A_t(X) = 1 \mid Y = 0, t\}. \tag{1}$$

The design seeks higher detection with an acceptable false-alert burden. These probabilities define the objective, not a solved constrained optimization problem: target attacks are unavailable during fitting, and the calibration parameter below does not certify a population constraint. The benchmark estimates class-specific rates on retained observations and macro-averages attack types, targets, and seeds as specified in Section 5. The local fitting budget N

is not the total number of target observations required. Reference and calibration observations are additional. Furthermore, the executed preprocessing pools the complete benign training roles of the loaded clients to estimate feature means and scales before restricting local detector fitting to T_N. This includes target training observations beyond the first N. Consequently, the experiment measures sensitivity to the local detector fitting budget conditional on this preprocessing and on additional benign reference and calibration data. It does not establish deployment after observing only N target rows.

Table 2: Data access in the executed benchmark. The target's retained benign roles follow chronological 40/20/20/20 proportions. N restricts local fitting within the training role, not total target data access.

Data role	Function
Target benign training (40%)	First N rows fit local detectors; the complete role contributes to pooled preprocessing.
Target benign reference (20%)	Constructs empirical target-relative evidence for peer and local scores.
Target benign calibration (20%)	Sets thresholds and calibrates the peer, anchor, and bank unions.
Target benign test (20%)	Measures held-out false-alert rates.
Target held-out attacks	Evaluates detection only; no target fitting, reference construction, or calibration.
Peer labeled development data	Trains supervised peer classifiers using benign and attack examples.

We denote the nominal operating parameter by α, the local reserve fraction by r, and the peer allocation by $B_P = (1 - r)\alpha$. The local allocation parameter is $B_L = r\alpha$. These parameters specify calibration operations; they are not bounds on the false-alert probability of future traffic. In particular, the implemented resolution rule can increase the sum of the component allocations above α, as detailed below.

The information regime assumes trusted benign target observations, trustworthy peer models, and compatible feature extraction. The benchmark prepares clients centrally with pooled preprocessing; its security and deployment boundaries are assessed in Section 8.

4. PACE methodology

Figure 1 separates the supervised peer route from the two target-local decisions. Algorithm 1 summarizes preparation and inference; the equations below specify the score and quantile conventions.

4.1. Target-relative evidence

For an expert score s, let $r_1, \ldots, r_n$ be that expert's scores on the target reference role, $r_{\max} = \max_i r_i$, and $d_R = \max\{r_{\max} - \text{median}_i\, r_i, 10^{-6}\}$. The implemented transformation is

$$E_R(s) = \begin{cases} -\log\left(\dfrac{1 + \sum_{i=1}^{n} \mathbf{1}\{r_i \geq s\}}{n+1}\right), & s \leq r_{\max}, \\ \log(n+1) + \log\left(1 + \dfrac{s - r_{\max}}{d_R}\right), & s > r_{\max}. \end{cases} \tag{2}$$

Ties are included in the upper-tail count. Values beyond the largest reference score receive an extrapolated value, rather than a saturated empirical rank. This extrapolation is a score convention, not an estimated tail probability with a distribution-free validity guarantee. For peer k, the evidence is $e_k(x) = E_{R,k}(m_k(x))$.

4.2. Blocks and thresholds

Let $M = |C|$ and let $b = 250$ be the configured block-size parameter. The code constructs $J = \max\{1, \lfloor M/b \rfloor\}$ consecutive blocks using a balanced array partition: their lengths differ by at most one, and earlier blocks receive any surplus rows. They are not necessarily blocks of exactly 250 observations. For example, 350 calibration observations form one block of length 350. Write $n_j = |C_j|$ and let $z_{j,(1)} \le \cdots \le z_{j,(n_j)}$ be ordered evidence in block j. For allocation $w \in (0, 1)$,

$$\tau_z(w) = \max_{1 \le j \le J} z_{j,(1+\lceil (1-w)(n_j-1) \rceil)}. \tag{3}$$

This is NumPy's higher empirical quantile followed by a maximum over blocks. Every decision uses a strict comparison $z(x) > \tau_z(w)$; equality does not alert. For a fixed threshold obtained this way, the observed exceedance fraction within each calibration block is at most w. This statement concerns the observations used to choose the threshold and does not extend to future data without additional assumptions.

4.3. Separately calibrated experts and union scaling

For a collection of L experts with allocation B, the default initial shares are $w_\ell = B/L$. Given multiplier λ, the union is

$$U_{B,\lambda}(x) = \mathbf{1}\{\exists \ell : z_\ell(x) > \tau_{z_\ell}(\lambda w_\ell)\}. \tag{4}$$

The calibrator measures this union on every calibration block. Starting at $\lambda = 1$, it doubles an upper search point until that point violates the block-rate criterion or reaches the allowed cap, returns immediately if the cap is feasible; otherwise it performs 20 bisections to retain a feasible lower point. The configured cap is 64. If the largest share times 64 is at least one, the implementation replaces the multiplier ceiling by $0.999/\max_\ell w_\ell$. If the initial union already violates B, the routine returns its initial thresholds; it does not claim to repair arbitrary supplied shares. PACE's equal-share initial unions satisfy the calibration criterion by the fixed-threshold count bound and the union bound.

The retained multiplier is a numerical search result on a stepwise calibration function. It should not be interpreted as a unique population optimum. The peer branch applies this routine to the K separate margin-evidence experts with allocation B_P. Separate thresholds avoid collapsing all peer evidence into one maximum before calibration. The common multiplier nevertheless couples the thresholds through the observed union, so an unstable peer can still affect their joint calibration.

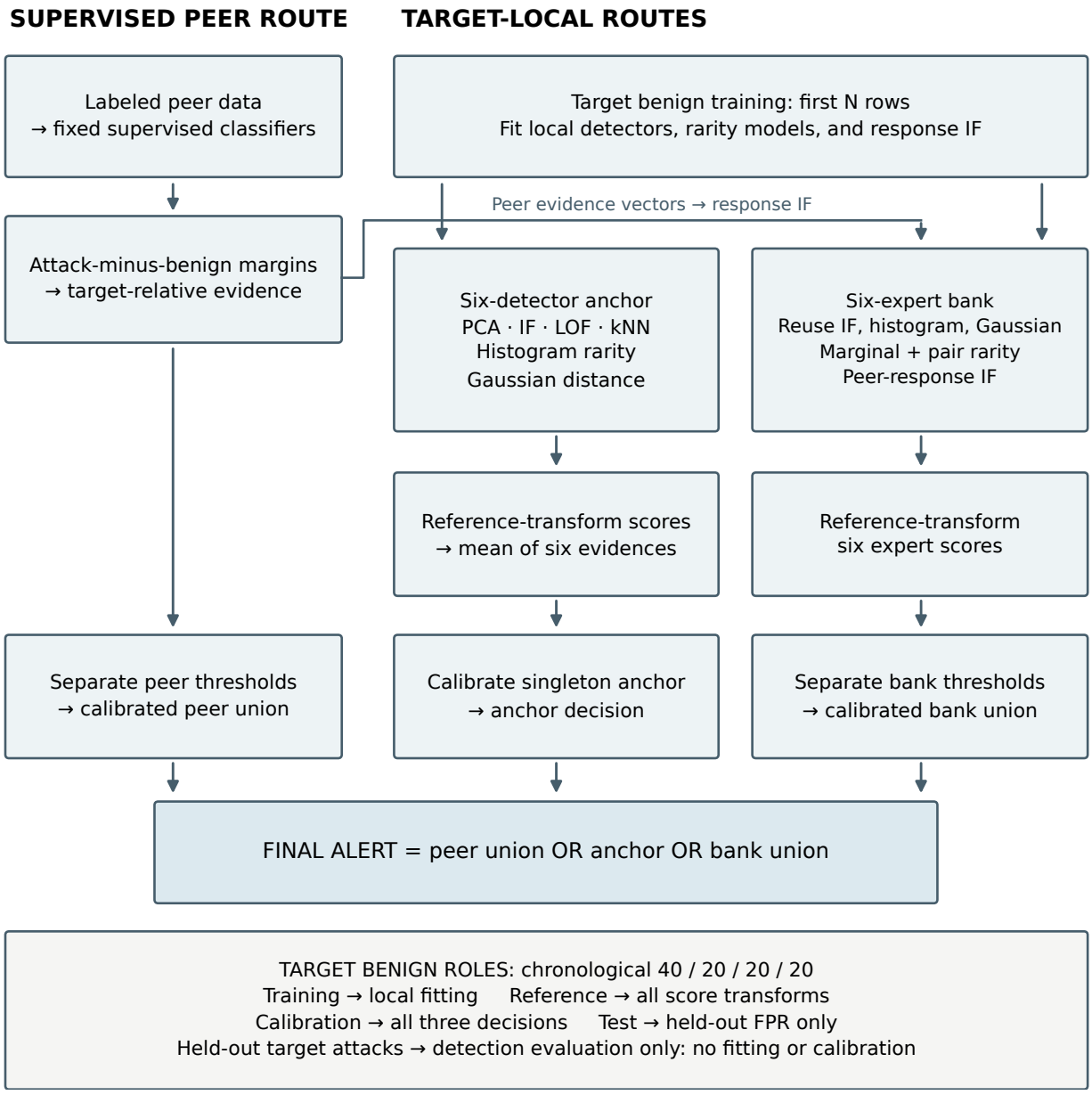


Figure 1: PACE's information flow and final decision. Target benign training, reference, calibration, and test roles are distinct. The benchmark standardizes features using pooled complete benign training roles before restricting local fitting to N rows. Target attack labels enter evaluation only. IF: isolation forest; LOF: local outlier factor; kNN: nearest-neighbor distance.

4.4. Anchor and complementary local bank

The anchor averages the reference-transformed scores of six local detectors: PCA reconstruction error, isolation forest, local outlier factor, mean nearest-neighbor distance, histogram rarity, and regularized Gaussian distance. Evidence is computed separately for each detector before averaging. The anchor is not transformed a second time. PCA retains 95% of explained variance and uses mean squared reconstruction error. Isolation forest [12] uses 100 trees, local outlier factor [13] uses 20 neighbors, and the distance score averages 10 neighbor distances from the fitted neighbor structure. Histogram scores sum negative log smoothed bin frequencies across features, with 20 equal-width training bins and additive-one smoothing. Values outside a feature's training range use its minimum fitted bin probability. The Gaussian score is a squared Mahalanobis distance using a pseudoinverse of the sample covariance plus $0.001I$.

The local bank contains six experts. Three reuse the anchor members' isolation-forest, histogram, and Gaussian evidence. Two score feature rarity, and one scores the joint response of the peers. These feature experts reuse fitted detector outputs; they are not three additional independently fitted copies.

The first rarity expert sums two-sided empirical tail evidence. For feature q, let $a_q(v)$ count training values at most v and $c_q(v)$ count training values strictly below v. Its raw score is

$$s_{\text{tail}}(x) = \sum_{q=1}^{d} \max\left\{ -\log \frac{a_q(x_q)+1}{N+1}, -\log \frac{N - c_q(x_q)+1}{N+1} \right\}. \tag{5}$$

This ECOD-inspired two-tail rarity score is labeled ECOD-sum in the implementation. It differs from original ECOD [25], which maximizes three aggregate scores formed from left, right, and skewness-selected tails. Here the featurewise larger negative-log tail is selected before summation; the equation specifies the implemented variant.

For pair co-occurrence, 150 ordered feature-index pairs are drawn with replacement. Self-pairs are discarded when more than one feature is available; duplicate pairs can remain, so 150 is a draw count rather than a guaranteed count of distinct retained pairs. Each feature is coded using its unique internal training quantiles at levels $1/8, \ldots, 7/8$, with left insertion at ties. Although these cut points give at most eight active codes, the implementation allocates a table of width $8 + 1 = 9$ in each dimension. If $h_{uv}(x)$ is the training count in the coded cell for retained pair (u, v), the score is

$$s_{\text{pair}}(x) = \frac{1}{|\mathcal{P}|} \sum_{(u,v)\in\mathcal{P}} -\log \frac{h_{uv}(x) + 0.5}{N + 0.5 \times 9^2}. \tag{6}$$

The reference transformation in Eq. (2) is applied to both rarity scores.

The response expert fits a 100-tree isolation forest to vectors $\rho(x) = (e_1(x), \ldots, e_K(x))$ for $x \in T_N$. It uses the negative forest score as its anomaly score and then applies Eq. (2) relative to its scores on R. Thus it models how the collection of foreign classifiers responds to target benign data. It neither trains a gating network nor changes the supervised peer models.

4.5. Resolution rule and final decision

With anchor share $a = 0.5$, the anchor allocation is

$$B_A = \begin{cases} aB_L, & aB_L \geq 1/b, \\ B_L, & aB_L < 1/b, \end{cases} \qquad B_B = (1-a)B_L. \tag{7}$$

The anchor is itself calibrated by the union-scaling routine as a singleton expert with allocation B_A. The bank is separately calibrated as a six-expert union with allocation B_B. In particular, the anchor threshold is not merely the unscaled quantile $\tau_A(B_A)$. Let the resulting decisions be U_P, U_A, and U_B. The final rule is

$$D(x) = U_P(x) \vee U_A(x) \vee U_B(x). \tag{8}$$

There is no final calibration pass over this three-way union.

At $\alpha = 0.01$ and $r = a = 0.5$, these allocations are $B_P = 0.005$, $B_A = 0.005$, and $B_B = 0.0025$. Their sum is 0.0125, rather than 0.01. An allocation-respecting alternative calibrates all local experts as a single union within the original total allocation. At $\alpha = 0.01$, this alternative attains the same pooled detection as PACE to the precision of the stored results, with nearly identical mean FPR (Appendix A). This sensitivity result indicates that the observed aggregate benefit does not require a measured detection advantage over the allocation-respecting alternative; it does not establish identical record-level decisions or a population FPR guarantee. At $\alpha = 0.05$, they are 0.025, 0.0125, and 0.0125, whose sum is 0.05. Overlap and integer count resolution can make the observed calibration union smaller than the allocation sum, but do not justify presenting Eq. (7) as a strict partition of α. The rule checks $1/b$, whereas the first possible nonzero exceedance for an individual higher quantile depends on its actual block length through $1/(n_j - 1)$. The resolution rule is therefore an implemented heuristic, not an exact test of every block's quantile resolution. The joint criterion permits at most $\lfloor Bn_j \rfloor$ alerts per block, whereas an individual unscaled higher quantile admits at most $\lfloor w(n_j - 1) \rfloor$ strict exceedances; these are different resolution limits.

The protected anchor preserves a separate local decision path but does not prove non-inferiority to the primary comparator. That comparator averages seven detector evidences, including an autoencoder omitted from the six-member anchor, and uses a different threshold procedure. Target-level regressions must be assessed empirically.

4.6. Training and computation

Each peer classifier has two ReLU hidden layers of widths 64 and 16 and two output logits. Training uses class-balanced cross-entropy, Adam updates, eight epochs, batch size 256, and learning rate 0.001. Target-local fitting uses the first N retained training rows. The feature-space isolation-forest and neighbor-based fitting sample is limited to 8000 rows; other local components have their own full-N fitting operations. Fitted score references and thresholds are fixed at inference: the evaluated rule has no online update.

The benchmark's shared local ensemble also trains and scores an autoencoder for the mean-ensemble comparator, even though its score is excluded from PACE's anchor. It would therefore be inaccurate to describe the executed campaign as requiring no local neural training. The detection architecture requires K peer forward evaluations plus local score computations; the response forest adds a model in K dimensions. A peer MLP contains $64d + 1138$ weights and biases for d input features. This algebraic parameter count is not a measurement of serialization size, communication traffic, latency, energy use, or deployment memory. None of those operational quantities is established by the reported detection results.

Algorithm 1. PACE preparation and inference

Input: fixed peer classifiers; target benign roles T_N, R, C; nominal α; $r = a = 0.5$; block parameter $b = 250$.

1. Evaluate each peer's attack-minus-benign margins on T_N, R, C; construct reference transforms $E_{R,k}$ by Eq. (2), including ties and beyond-reference extrapolation.
2. Fit the local detectors and rarity models on T_N, and the response IF on $(e_1, \ldots, e_K)(T_N)$. Transform each local score against its scores on R. Form the six-evidence mean anchor and six-expert bank.
3. Partition C into $\max(1, \lfloor |C|/b \rfloor)$ balanced consecutive blocks. For each expert, use the maximum block higher quantile in Eq. (3).
4. Set $B_P = (1-r)\alpha$ and $B_L = r\alpha$. Compute B_A, B_B with the resolution rule in Eq. (7).
5. Apply CalibrateUnion to the K peers with B_P, the singleton anchor with B_A, and the six bank experts with B_B. Store all reference transforms and thresholds.
6. For a query x, compute the same frozen evidences. Alert if any peer exceeds its threshold, the anchor exceeds its threshold, or any bank expert exceeds its threshold. All comparisons are strict $>$; no final recalibration is applied.

CalibrateUnion(Z, B): initialize equal shares $w_\ell = B/L$ and thresholds at $\lambda = 1$. Use cap $c = 64$, replaced by $0.999/\max_\ell w_\ell$ if $64 \max_\ell w_\ell \geq 1$. Return the initial thresholds if their worst block union rate exceeds B. Otherwise test $\min(2, c)$ and double up to c while feasible; return immediately if c is feasible. Once an infeasible upper point is found, perform 20 bisections, retaining the feasible lower point. Return its thresholds.

5. Experimental methodology

5.1. Evaluation units and eligibility

The evaluation contains six corpora organized into seven study strata: IoT-23 [26] (11 targets), TON_IoT [27] (9), Gotham [28, 29] (7), CTU-13 [30] (12), N-BaIoT [15, 31] (9), and UNSW IoT attack flows [32] G1 and G2 (5 each). These 58 units are the eligible benchmark targets, not a random sample of deployments. Appendix B records the exact input archives or subsets, authoritative source links, adapters, and verified access conditions. Each stratum uses its own compatible feature schema. Peer transfer occurs within the prepared pool for that stratum, rather than across corpora.

Reported-target membership is predefined by corpus-specific client manifests; eligibility is then checked within those manifests. A target must have at least one eligible attack type, at least 350 calibration observations, and at least four eligible peers. The configured minimum number of retained rows for an attack type is 600 by default, reduced to 400 for IoT-23, TON_IoT, and CTU-13, and to 30 for the two UNSW strata. Eligibility therefore depends partly on the availability of evaluation attacks even though attack labels do not enter the target detector's fitting or calibration. This distinction matters when interpreting benign-only deployment claims.

The census records 769–80,000 benign training-role rows per target. Peer counts range from four in each UNSW group to eleven in CTU-13. Gotham's seven reported targets each have seven peers, indicating that reported-target and peer-pool membership must not be assumed identical. Corpus-specific preprocessing and retained sampling precede the experimental role

split. Chronological ordering of retained observations is not equivalent to evaluation of an untouched continuous capture stream.

Table 3: Prepared target units and feature dimensions. Counts describe eligible reported targets; peer pools can be larger. Features and sampling are adapter-specific.

Stratum	Evaluation unit	Targets	Features
IoT-23	Capture/scenario	11	36
TON_IoT	Destination host	9	29
Gotham	Device capture	7	37
CTU-13	Botnet scenario	12	38
N-BaIoT	Device directory	9	115
UNSW G1	Device	5	15
UNSW G2	Device	5	13

Table 3 distinguishes the evaluation units and input dimensions. TON_IoT's selected destinations include routers, servers, and conventional hosts, so the collection should not be read as 58 consumer IoT devices. Each adapter applies a signed logarithmic feature transformation before pooled standardization. Sampling and eligibility operate on retained rows. IoT-23 caps each raw detail bucket at 200,000 benign or 60,000 malicious rows; TON_IoT caps each host at 150,000 benign rows and each host–attack group at 60,000. IoT-23 additionally thins detail groups larger than 20,000 rows to 2% within million-row chunks, only for files larger than 1.5 billion bytes. TON_IoT thins host–label groups larger than 20,000 rows to 3% within two-million-row chunks, using an intermediate 60,000-row reservoir. Both thinning counts are rounded down. CTU-13 retains at most 200,000 normal and 60,000 attack rows per scenario and discards Background traffic. Gotham retains at most 200,000 benign rows and 40,000 per attack type. N-BaIoT applies stable exact-row deduplication within files, and its source order substitutes for timestamps. The UNSW inputs use annotated minute-level flow counters and have no adapter reservoir cap.

Gotham's features require particular caution. Five features summarize all packets within each complete second, before label filtering and reservoir sampling. A retained benign row can therefore contain aggregate information about attack or unlabeled packets in that second, and later packets contribute to earlier rows' features. Role boundaries that divide a second do not separate the raw observations contributing to these aggregates. This dependency does not establish that attack labels entered model fitting, but it prevents interpreting the benchmark as immediate causal packet-stream detection.

5.2. Role separation and fitting budgets

The benign observations of each prepared client are ordered by timestamp or recorded source order, using a stable ordering. Split boundaries are the floors of the cumulative fractions 0.4, 0.6, and 0.8, producing training, reference, calibration, and test roles. The first N training-role observations fit the local detectors. Insufficient rows cause the corresponding budget to be ineligible; observations are not borrowed from later roles to meet it.

For each eligible attack type, the ordered retained observations are divided at the midpoint. Up to the first 10,000 observations of the first half supply peer-development examples; up

to the first 10,000 observations of the second half supply evaluation examples. A target's supervised model is omitted when that client becomes the monitored target. The eligible foreign peers may contain development examples with attack labels also present at the target. Across the executed evaluation, only 11 of 258 target–attack-type pairs have labels absent from every eligible peer's development data. This label-level overlap does not imply identical attack instances or behavior, and the experiment should be interpreted as detection without target attack labels rather than systematic held-out-family detection. These chronological splits are applied separately within clients and labels; they do not impose a common wall-clock cutoff across the network.

The pooled standardizer is fitted before budget-specific local fitting, using complete benign training roles of the loaded clients. Reference and calibration roles remain separate from benign test rows in the decision pipeline. Nevertheless, this preprocessing means that the total target data available to the experiment exceeds N, as explained in Section 3. Peer classifiers also use this common scaling, so target-derived benign statistics influence their input representation. No claim of fitting the entire system from 100 or 400 total target observations is made.

Four analyses avoid mixing changes in budget with changes in eligible targets. The universal primary analysis compares $N = 100$ and 400 on all 58 targets. Universal robustness adds $N = 750$ on those same targets. Broad scaling compares $N = 100$, 400, and 2000 within the 47-target cohort supporting 2000 rows. The high-resource analysis compares $N = 100$, 400, 2000, and $20,000$ within the 15-target cohort supporting the largest budget. Budget effects are paired within their specified cohort.

5.3. Configuration and comparison rules

Both $\alpha = 0.01$ and 0.05 are evaluated. The reserve fraction is 0.5, the block-size parameter 250, the multiplier cap 64, and the bisection count 20. The learning seeds are 6, 7, and 8. Component seeds are derived by target identity and component domain; these runs vary learning randomness while keeping data roles fixed. The configuration's single default alpha is not the scope of the campaign: the protocol matrix explicitly includes both levels.

The primary comparator is PACE with a mean-ensemble local branch. Both methods use the same supervised peer architecture, target-relative peer evidence, peer calibration, target roles, learning-seed schedule, and evaluation cohort. The comparator thresholds the average evidence of seven local detectors at the full local allocation. The seventh detector is an autoencoder with hidden widths 20 and 10, a symmetric decoder, 30 training epochs, batch size 256, and learning rate 0.001. This comparator differs from the six-member anchor in both composition and calibration, so the primary comparison is an assessment of the complete local-branch replacement.

Two recorded baselines also retain this seven-detector local branch. The max-fusion baseline transforms peer attack probabilities relative to target reference probabilities, takes their maximum, and calibrates that fused value once. Its beyond-reference transformation differs from the margin transformation in Eq. (2): it adds the logarithm of the probability-to-reference-maximum ratio to $\log(n + 1)$. The Bonferroni baseline uses equally shared margin-evidence peer thresholds with multiplier fixed to one. These baselines should not be described as peer-only detectors.

Separate peer-mean and peer-maximum ablation arms fuse PACE's margin evidence while retaining the PACE local bank. These arms distinguish the effect of fused peer calibration from the probability-based recorded max-fusion baseline. The no-local-channel arm allocates the entire nominal parameter to the peer union, whereas the reported peer branch inside PACE receives only its reserved peer share. The local-only arm uses the seven-detector mean at the entire nominal parameter. Neither standalone arm is identical to a branch extracted from the final combined rule.

Additional arms remove the feature, rarity, or response family; remove the anchor; combine anchor and bank into a single weighted union; fit the response expert on margins; or reduce the local reserve to 0.25. Every arm uses the same relevant data roles and seed schedule. Appendix A reports PACE, all 12 executed internal study arms, and both recorded baselines, including the alternatives with adverse or inconclusive contrasts. These are internal architectural comparisons. The campaign does not provide implementations of every contemporary federated, transfer, or calibrated intrusion detector, so literature positioning is not an empirical superiority comparison with those methods.

5.4. Metrics and aggregation

For target t, attack type h, and seed s, detection is the fraction of held-out attack observations that alert. The target detection rate averages these fractions equally across its attack types and then across the three seeds. The pooled detection rate averages targets equally. Large captures therefore do not dominate through their row counts, although strata with more targets have greater pooled weight. False-alert rate is the fraction of benign test observations that alert, averaged over seeds within target and then over targets for its pooled mean.

The primary paired difference subtracts the comparator's target detection from PACE's. Intervals use 10,000 bootstrap replicates with seed 314159 and confidence level 0.95. Targets are sampled with replacement within corpus groups identified by the dataset enum, preserving each group's size, and their contributions are weighted by target count. The interval endpoints are linear empirical quantiles of the bootstrap means. The two UNSW study strata belong to one corpus group for this resampling. Seeds and attack observations are not resampled as independent targets.

This procedure conditions on the observed corpora and the seed-averaged target effects. It does not account fully for shared peer models, common captures, cohort selection, or dependence between target units. Intervals are descriptive evidence about this benchmark; they are not uncertainty bounds for arbitrary new IoT populations. Multiple ablations and subgroup analyses are exploratory, without a claim of simultaneous family-wise coverage.

Sweep-adjusted primary comparisons use multipliers $\{0.55, 0.75, 1, 1.4\}$ at $\alpha = 0.01$ and $\{0.4, 0.75, 1, 1.4\}$ at $\alpha = 0.05$, scaling both branch allocation parameters. PACE's complete rule, including its resolution condition, is evaluated at each stored point. Primary reporting interpolates within the realized-rate sweep and clamps outside its endpoints, separately for each seed. This procedure approximates a comparator operating point; it does not guarantee exact matching. The false-alert distribution reports quantiles over target seed means, while acceptance additionally examines exceedance among target–seed cells. These denominators are distinct.

5.5. Internal protocol checks and reproducibility

The primary engineering acceptance rule permits no target loss exceeding 0.05 and permits at most 3% of targets to lose more than 0.01. It also requires positive lower bounds for nominal and sweep-adjusted pooled gains; limits mean false-alert increases to 0.004 or 0.002 at the two alpha levels; limits the increase in target–seed exceedance above 2α to 0.03; and checks excluded-TON_IoT, learning-seed, and leave-one-stratum-out summaries. These conditions describe the repository's decision rule. They do not constitute a formal non-inferiority hypothesis test, a registered clinical-style margin analysis, or external validation.

The outcome-independent power table is a planning calculation under a normal approximation over assumed paired standard deviations and a meaningful effect of 0.02, using level 0.05 and power 0.8. It is not an observed-power validation of the results. At 58 targets, for example, assumed standard deviations of 0.05 and 0.10 produce minimum detectable effects of 0.018393 and 0.036787. The conclusions rely on the actual paired outcomes and their limitations rather than on selecting a favorable planning assumption.

The benchmark cohort also informed architectural development. An earlier failure on IoT-23 capture 35-1 motivated the anchor resolution rule. The project specification identifies learning seeds 6–8 as unused for final design selection; their evaluation measures stability under learning randomness on the reused cohort rather than independence from the development population. The Git record first includes the architecture, comparison-arm code, configuration, and acceptance checks together on 10 October 2026; frozen result tables were added on 11 October and the manuscript on 12 October (dates in Africa/Casablanca). These commit dates establish repository chronology, not when private outcomes were first inspected. **Author confirmation required:** exact architecture and acceptance-criterion freeze dates, and whether any comparison arms were specified after primary outcomes were inspected.

Numerical summaries use the existing campaign tables or deterministic recalculation from the same frozen outputs. No models were retrained and no scientific campaign was rerun during manuscript preparation. The 110 campaign artifacts contain 1824 target–seed records, including 408 identical repeats at shared resource-cohort budgets; the 1416 unique $(\alpha, N, \text{stratum}, \text{target}, \text{seed})$ keys are counted once per analysis. The manuscript reporting verifier and figure-only generation commands are documented in `manuscript/REPRODUCIBILITY.md`. Provenance checks establish artifact identity and consistency; they do not extend the evaluation to other target populations.

6. Experimental results

6.1. Primary detection gains and mean false-alert tradeoffs

Across 58 targets and three learning seeds, PACE improves target-macro detection relative to the mean-ensemble local comparator in all four primary configurations (Table 4). At nominal $\alpha = 0.01$, the gains are 2.61 and 2.83 percentage points with 100 and 400 fitting rows. At $\alpha = 0.05$, the corresponding gains are 2.34 and 2.32 points. All four corpus-stratified paired bootstrap intervals exclude zero. Because the two methods share the same peer branch, these comparisons measure the benefit of replacing the complete local branch.

The $\alpha = 0.05$ configurations also show lower mean realized false-alert rates for PACE. At 100 rows, detection rises from 87.13% to 89.48%, while mean FPR falls from 5.74% to 5.55%. At 400 rows, detection rises from 87.36% to 89.68%, while mean FPR falls from 6.44% to

5.43%. At $\alpha = 0.05$ and $N = 400$, two UNSW G2 targets account for approximately 0.0080 of the 0.0101 mean-FPR reduction. Excluding these targets, PACE still reduces mean FPR by 0.002107, with a paired interval of $[0.000655, 0.003869]$ for the reduction magnitude. Across all 58 targets, 32 have lower FPR under PACE, 23 have higher FPR, and three are unchanged. These aggregate improvements show that the added detections at these nominal settings do not require a higher mean benign alert rate. They describe target-macro outcomes at the same nominal parameter; individual targets need not improve on both measures.

Table 4: Primary comparison on 58 targets at the same nominal α, rather than exactly matched realized FPR. Detection is averaged over attack types, learning seeds, and targets; mean FPR is averaged over seeds and targets. Intervals are 95% corpus-stratified paired bootstrap intervals. Intervals condition on the observed corpora and seed-averaged targets; shared peers and new deployment populations are not resampled. All entries are proportions; gains are computed before rounding.

α	N	PACE TPR	Comparator TPR	Gain	95% interval
0.01	100	0.833405	0.807318	0.026086	[0.011323, 0.043346]
0.01	400	0.838987	0.810681	0.028306	[0.012339, 0.047375]
0.05	100	0.894767	0.871348	0.023418	[0.005777, 0.043968]
0.05	400	0.896837	0.873619	0.023219	[0.003116, 0.045968]

α	N	PACE mean FPR	Comparator mean FPR	FPR difference	95% interval
0.01	100	0.033992	0.030516	0.003476	[0.000621, 0.008522]
0.01	400	0.032369	0.030781	0.001588	[0.000537, 0.003249]
0.05	100	0.055453	0.057437	-0.001984	[-0.003909, -0.000189]
0.05	400	0.054337	0.064407	-0.010070	[-0.021626, -0.001486]

At nominal $\alpha = 0.01$, detection also improves, although mean FPR increases by 0.003476 and 0.001588 at the two budgets. These gains therefore accompany an alert cost. The conditional interpretation of the paired intervals and the target-macro estimator follow Section 5.

A simpler alternative allocates the entire nominal operating parameter to the supervised peer union and removes the local branch. This peers-only rule achieves detection rates of 0.812411 at $\alpha = 0.01$ and 0.883945 at $\alpha = 0.05$. PACE improves on these means by 2.10–2.66 and 1.08–1.29 percentage points, respectively, although the paired detection intervals include zero. At $\alpha = 0.05$, PACE also reduces mean realized FPR relative to peers-only by 0.010774 and 0.011890 at fitting budgets of 100 and 400 rows. These comparisons complement the primary local-branch replacement experiment without establishing statistically resolved detection superiority over the peers-only rule.

6.2. Peer-wise calibration versus fused peer evidence

The peer-fusion arms keep PACE's entire local branch, including its anchor and expert bank, fixed. PACE exceeds mean-fused peer detection by 0.073437 and 0.055978 at $\alpha = 0.01$ and by 0.052459 and 0.044723 at $\alpha = 0.05$, for $N = 100$ and 400, respectively. Against maximum-fused peers, the corresponding differences are 0.045555, 0.036390, 0.030484, and 0.031644. All eight paired descriptive detection intervals exclude zero (Appendix A).

These nominal detection advantages accompany higher mean realized FPR. Mean-fused peers yield FPRs of 0.026093 and 0.024896 at $\alpha = 0.01$, and 0.047157 and 0.044361 at $\alpha = 0.05$. Maximum-fused peers yield 0.025120, 0.023855, 0.046431, and 0.043049 in the same order, compared with PACE's 0.033992, 0.032369, 0.055453, and 0.054337. The comparisons support retaining separate peer decisions at the tested nominal settings, with an alert-rate cost; they neither establish exact equal-FPR superiority nor isolate peer drowning as the cause.

6.3. Which components supply complementary detections?

Removing the rarity family costs 0.008626 and 0.011977 in pooled detection at $\alpha = 0.01$, compared with 0.003413 and 0.002013 for removing feature experts, and 0.002008 and 0.002796 for removing the response expert. At $\alpha = 0.05$, the rarity costs remain 0.006020 and 0.010403; feature costs are only 0.000127 and 0.000344. The response expert's costs are 0.003118 and 0.001269. Rarity-based evidence thus makes the largest measured contribution among the removed expert families in all four primary configurations.

Ablation changes also recalibrate the remaining union. They are not additive decompositions of the gain and cannot be summed to assign causal importance. Replacing target-relative peer responses by raw margins changes pooled detection by only 0.001101–0.002879 across primary settings. Several corresponding intervals include zero. The evidence for the particular response representation is therefore weaker than the overall comparison with the mean-ensemble local branch.

The anchor has a tradeoff. Removing it also reallocates the entire local budget to the expert bank; the ablation therefore changes both the decision structure and the bank allocation. At $\alpha = 0.01$, the anchor-containing rule benefits IoT-23 capture 35-1, while other targets, including TON_IoT host 192.168.1.190, favor the bank-only alternative. The pooled intervals include zero. At $\alpha = 0.05$, the bank without the anchor detects more: its rates are 0.897598 and 0.898372, versus PACE's 0.894767 and 0.896837. Its false-alert rates are also higher, at 0.060458 and 0.064650. A single local union gives identical detection to PACE at $\alpha = 0.01$ to the precision of the stored table; at $\alpha = 0.05$ its differences are small and its false-alert rates higher. These outcomes characterize an architectural and allocation tradeoff rather than an isolated causal effect of the anchor, with the anchor-free bank favoring detection at the larger nominal parameter.

Within the final PACE rule, the attack-detection decomposition distinguishes alerts unique to the local branch from detections shared with peers. At $\alpha = 0.01$ and 400 rows, local-exclusive detection is 0.147434 in TON_IoT and 0.120957 in IoT-23, but only 0.000042 in N-BaIoT. Under the final OR rule, a local alert adds detection only when peers miss the attack. These local-exclusive coverage fractions identify the opportunity for complementary detection; they are not gains over the mean-ensemble comparator or results for a separately calibrated local-only detector.

6.4. Where are the gains concentrated?

TON_IoT has the largest stratum mean gains: 0.117448 and 0.126729 at $\alpha = 0.01$, and 0.120780 and 0.104447 at $\alpha = 0.05$, for 100 and 400 fitting rows respectively. Removing its nine targets reduces the pooled gains to 0.009306, 0.010228, 0.005536, and 0.008299. Their intervals are $[0.001808, 0.019526]$, $[0.000995, 0.024240]$, $[0.000090, 0.014343]$, and $[0.000143, 0.022250]$. Excluding TON_IoT, the four primary pooled gains remain positive, ranging from 0.55 to

1.02 percentage points. Three of the four corresponding interval lower bounds are below 0.001. The three largest contributing targets account for approximately 60–89% of the summed gains, depending on the configuration. As an exploratory sensitivity analysis, removing the five largest-gain targets still leaves positive pooled gains of approximately 0.11–0.63 percentage points. These removals are selected by observed outcomes.

N-BaIoT is already near complete detection and changes by approximately 10^{-5} or less. Gotham gains range from 0.001950 to 0.005907. UNSW G2 improves by 0.009251 and 0.005548 at $\alpha = 0.01$, but changes by -0.001420 and -0.002037 at $\alpha = 0.05$. Several stratum intervals include zero. The median target gain is zero in many strata, so a positive mean should not be paraphrased as an improvement for most devices.

IoT-23's primary mean gains range from 0.017582 to 0.029670, but its target-level pattern is concentrated rather than uniform. At $\alpha = 0.01$ and 400 rows, CTU-13's mean gain is 0.003156 with interval $[-0.007871, 0.017066]$. Scenario 12 loses 0.032595, the only primary target loss exceeding 0.01. No primary target loses more than 0.05. Most of the seed-averaged loss for CTU-13 scenario 12 at $\alpha = 0.01$ and $N = 400$ occurs in seed 7, where the peer union detects only 0.189 of evaluated attacks and the comparator's local route recovers more detections. At $N = 100$, the same seed favors PACE. This localizes the regression to a particular seed and fitting-budget configuration without identifying a detector-level cause.

Using a descriptive tolerance of 0.001, at $\alpha = 0.01$ and $N = 400$, PACE improves 21 targets, is effectively unchanged on 36, and regresses on one. At $\alpha = 0.05$ and the same budget, the corresponding counts are 15, 41, and two. The targets share peer models, so these counts describe the observed benchmark rather than independent deployment outcomes.

Figure 2 displays the concentration of the gains and the negative tail without aggregating targets into stratum means.

6.5. Are false alerts compatible with the nominal parameter?

Table 5 gives the distribution of target false-alert rates after averaging over learning seeds. At $\alpha = 0.01$, PACE's mean is 0.033992 at 100 rows and 0.032369 at 400 rows, while the medians are 0.006485 and 0.006419. The large difference between means and medians indicates substantial heterogeneity. The worst target averages are 0.669005 and 0.670842. About 26% of targets exceed twice the nominal parameter in both settings. A nominal value of 0.01 therefore cannot be interpreted as a one-percent test-time false-alert bound.

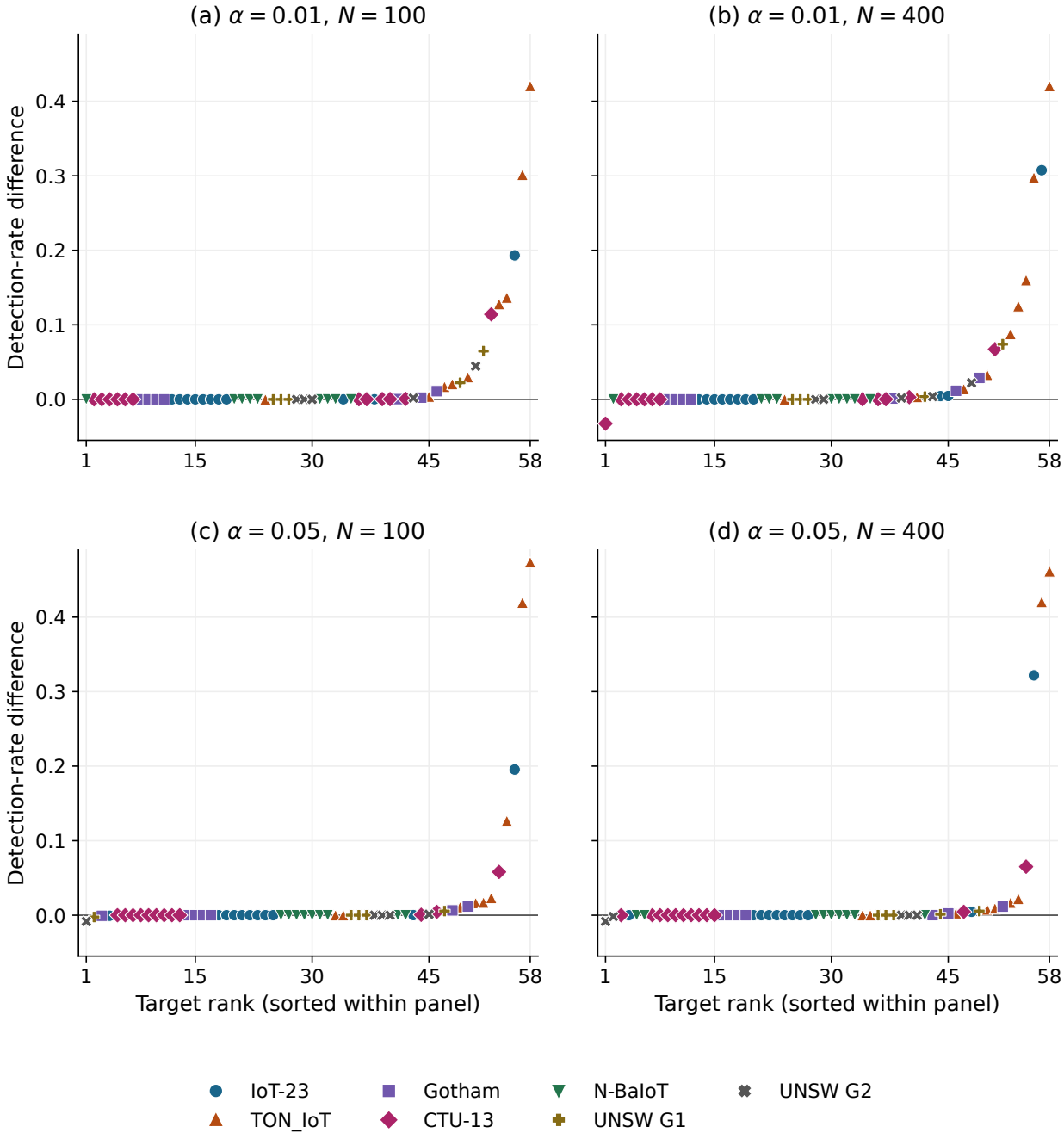


Figure 2: PACE minus comparator detection for all 58 targets per panel. Targets are independently sorted within each panel, so ranks do not identify the same target across panels. Colors and marker shapes identify strata; negative gains remain visible. Differences are proportions, not percentage points.

The comparator has the same qualitative failure: at $\alpha = 0.01$, its mean rates are 0.030516 and 0.030781, and its worst-target rates are 0.670174 and 0.670842. Calibration on earlier benign data therefore does not deliver uniform nominal control on the later benign role. The recorded outcomes do not isolate temporal change, sample selection, or score extrapolation as its cause. The low-alpha allocation exception in Section 4 also applies before any test-distribution change.

Read-only diagnostics reveal associations between excessive false alerts and changes in benign traffic across data roles. In IoT-23 capture 7-1, approximately 66% of benign test observations fall outside the calibration range of the `orig_ip_bytes` feature. Higher false-alert rates are also associated with smaller calibration sets across the benchmark, although corpus and sample-size effects are confounded. These observations identify useful diagnostic directions without isolating a responsible detector or proving that a larger calibration set would correct the problem.

At $\alpha = 0.05$, PACE's means are 0.055453 and 0.054337; the corresponding 90th percentiles are 0.162115 and 0.159692. The worst value remains 0.671844. Even where the pooled average approaches the nominal parameter, the upper tail is unsuitable evidence for uniform operational control. A row-level false-alert rate is also not an incident-level alert burden: aggregation, duplicate suppression, analyst capacity, and the benign-to-attack prevalence of a deployed stream were not measured.

Figure 3 shows the complete cumulative distributions, including the long upper tails that are obscured by pooled means.

6.6. How much remains after operating-point adjustment?

The stored budget sweep yields primary comparator-adjusted gains of 0.024659, 0.025735, 0.022176, and 0.020718 for the four settings in Table 4. The four intervals, in the same order, are $[0.009682, 0.042034]$, $[0.009690, 0.044983]$, $[0.005135, 0.042086]$, and $[0.002056, 0.042482]$. These estimates remain positive, but their interpretation requires the actual interpolation procedure.

For each target and learning seed, the reporting code sorts PACE's four sweep points by realized benign false-alert rate and linearly interpolates detection at the comparator's rate. When that rate falls outside the observed sweep, the calculation uses the nearest endpoint. It then averages the resulting differences over seeds and targets. Consequently, these figures are clamped sweep-adjusted comparisons, not evidence that every detector was evaluated at precisely the same false-alert rate. The primary calculation also differs from a separate max-fusion reporting routine that constructs a monotone envelope and rejects uncovered operating points. Claims about primary matched comparisons must use the former definition.

Table 5: Distribution of PACE's target-level realized false-alert rates, averaged over three learning seeds. Exceedance columns are proportions of targets, not proportions of individual benign rows or target–seed cells.

α	N	Mean	Median	90th pct.	Worst	$> \alpha$	$> 2\alpha$
0.01	100	.033992	.006485	.062058	.669005	.413793	.258621
0.01	400	.032369	.006419	.067903	.670842	.396552	.258621
0.05	100	.055453	.019118	.162115	.671844	.224138	.137931
0.05	400	.054337	.018427	.159692	.671844	.241379	.137931

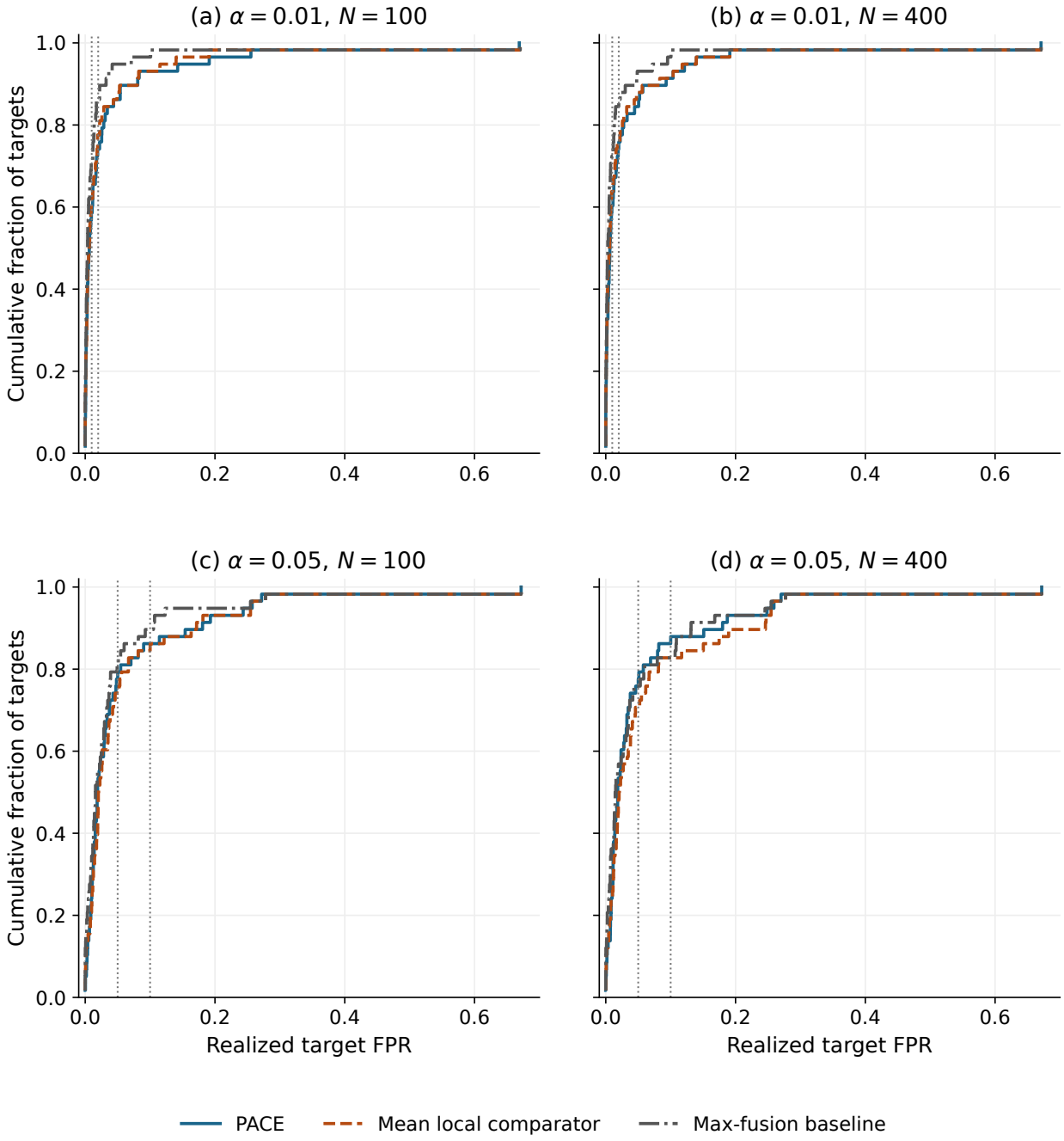


Figure 3: Empirical cumulative distributions of benign test FPR for all 58 targets, averaged over three seeds. Line styles distinguish PACE, its mean-local comparator, and the recorded max-fusion baseline (which also includes a mean-local branch). Vertical dotted guides mark nominal α and 2α, not demonstrated bounds.

The narrow four-point sweep and endpoint handling limit how much this analysis can resolve operating-point confounding. Of 174 target–seed cells per setting, the comparator rate falls below or above the PACE sweep in 47/28 cells at $\alpha = 0.01$, $N = 100$, and 49/20 cells at $N = 400$. At $\alpha = 0.05$, these counts are 12/24 and 9/19. Thus 75, 69, 36, and 28 comparisons respectively use endpoints outside the covered range. These counts are derived from the stored campaign sweeps; they make explicit how often the reported adjustment cannot actually match the comparator rate.

When the comparator's realized FPR falls below the evaluated PACE sweep, endpoint clamping uses PACE detection measured at a higher FPR. When the comparator's rate exceeds the sweep, clamping uses detection measured at a lower FPR. Consequently, these uncovered operating points do not identify the detection difference at exactly equal FPR. The sweep-adjusted results provide an approximate sensitivity analysis of the nominal comparison, rather than an independently measured matched-FPR superiority result.

6.7. Does more local fitting data reliably help?

On the same 58 targets, increasing the fitting budget from 100 to 400 rows changes PACE detection by 0.005582 at $\alpha = 0.01$, with interval $[-0.003760, 0.015387]$, and by 0.002070 at $\alpha = 0.05$, with interval $[-0.004456, 0.008690]$. Neither interval excludes zero. The local branch alone gains much more coverage, but much of that coverage overlaps detections already supplied by peers. The combined rule need not improve proportionally to the local branch.

At 750 rows, the universal cohort retains gains over the comparator of 0.024276 and 0.022614. However, at $\alpha = 0.05$, the interval for the gain excluding TON_IoT includes zero. The 2000-row analysis contains only the 47 targets with enough training-role observations. Its gains over the comparator are 0.019624 and 0.007358; both sweep-adjusted intervals include zero. At $\alpha = 0.05$, the gain excluding TON_IoT is negative, -0.000370, with interval $[-0.000712, -0.000074]$.

The 20,000-row cohort contains 15 targets. Its nominal gains are 0.005329 at $\alpha = 0.01$ and -0.001311 at $\alpha = 0.05$, with intervals including zero. Within this cohort, increasing the budget from 100 to 20,000 rows at $\alpha = 0.05$ reduces PACE detection by 0.028296, with interval $[-0.032827, -0.023766]$. In the 15-target high-resource cohort, this decrease is concentrated in TON_IoT host 192.168.1.190, whose detection changes by -0.362268. Excluding this host, the remaining 14 targets have a mean change of -0.004441, with a paired interval of $[-0.009296, 0.000412]$. Because TON_IoT contributes only one target to this cohort, the corpus-stratified bootstrap holds its contribution fixed. The narrow pooled interval therefore describes the observed cohort under this resampling procedure and does not establish a general decrease in detection from using more local fitting data. The results do not support monotonic improvement with additional local fitting data. Comparisons across the 58-, 47-, and 15-target cohorts would confound fitting budget with target composition and are not used as estimates of budget effects.

Figure 4 summarizes comparator gains within each fixed resource-eligible cohort.

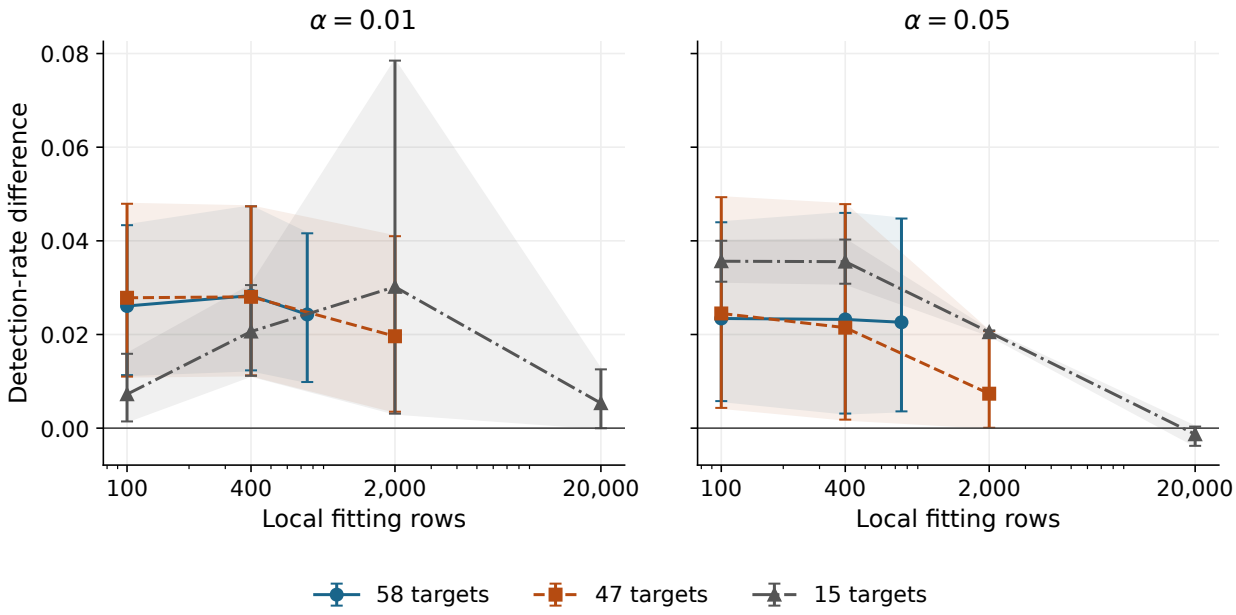


Figure 4: Detection gains over the mean-local comparator within three fixed cohorts: 58 targets through $N = 750$, 47 through $N = 2000$, and 15 through $N = 20,000$. Markers and error bars show evaluated budgets and 95% corpus-stratified bootstrap intervals. Lines and shading are visual guides only. Compare budgets within a cohort; between-cohort differences also reflect target composition. Intervals condition on the observed resource-eligible cohorts; the 15-target cohort contains a singleton TON_IoT contribution that is fixed under corpus-stratified resampling.

6.8. Seed stability and engineering acceptance

Every learning seed has a positive primary pooled gain, ranging from 0.015977 to 0.039551 across settings. This consistency is narrower than generalization to new targets: all three seeds reuse the same data and splits. Excluding TON_IoT, seed 6 has a gain of -0.000195 at $\alpha = 0.05$ and 400 rows. Every primary leave-one-study-stratum-out pooled gain is positive, but this sensitivity check still operates on the same collection of corpora and targets.

The primary cells satisfy the repository's engineering acceptance criteria, which permit a limited fraction of harmed targets, bound changes in realized false alerts, and require positive aggregate summaries. This is not a formal statistical non-inferiority test or independent external validation. In particular, the observed scenario-12 regression remains a regression despite satisfying the aggregate acceptance rule. The larger-budget non-gating analyses contain failures of individual criteria and are reported as such, rather than being used to extend the primary verdict.

7. Discussion

7.1. Complementary local coverage

The principal benefit of the heterogeneous local branch is complementary coverage. Under the final OR rule, locally detected attacks improve the combined decision only when the supervised peer union misses them. This explains why the gains depend on the availability of peer-missed attacks and why improved standalone local detection need not translate into

proportional end-to-end gains. Local-exclusive coverage at $\alpha = 0.01$ and $N = 400$ reaches 14.74% in TON_IoT and 12.10% in IoT-23, whereas it is nearly zero in N-BaIoT.

The rarity-family removals produce the largest detection losses among the tested expert-family ablations in all four primary configurations. This supports the usefulness of marginal and pairwise feature-rarity evidence alongside the shape- and neighborhood-based detectors. The smaller response-expert effects indicate that joint peer-response modeling supplies additional, but less extensive, coverage. Because each removal recalibrates the remaining union, the losses characterize complete ablated rules rather than an additive attribution to individual experts.

The protected anchor preserves an averaged local decision path alongside sparse expert alerts. Its value is an operating tradeoff: at $\alpha = 0.05$, removing it raises both detection and false alerts. The primary comparison also changes the comparator's seven-member composition and thresholding, so the overall gain belongs to the complete local-branch replacement.

7.2. Retaining peer evidence separately

Peer-wise calibration gives each classifier its own target-relative threshold. Unlike a threshold on a fused maximum, a high threshold for one peer need not directly exclude another peer's smaller evidence. This structure addresses the proposed peer-drowning failure mode while retaining dependence through the common union multiplier; the aggregate outcomes do not isolate that mechanism.

The fixed-local-branch comparisons in Section 6.2 support retaining separate peer decisions. The peer-wise architecture trades higher detection for higher mean FPR at the tested nominal settings; the clamped sweep estimates provide additional, approximate operating-point sensitivity evidence.

The shared multiplier uses overlap between calibration alerts to relax thresholds while checking their union on every block. It does not require independent peer alerts. It also couples the retained thresholds, since changing one expert's calibration alerts can change the multiplier for all peers. The empirical evidence transformation and union search retain the calibration interpretation specified in Section 4.

7.3. Variation across targets and fitting budgets

TON_IoT offers the largest observed gain and substantial local-exclusive coverage. IoT-23 also benefits on average, although its gains are concentrated in a subset of targets. N-BaIoT's near-complete peer coverage leaves little room for the local path to improve the final union. CTU-13 has small and uncertain stratum gains and includes the scenario-12 regression. These outcomes locate the empirical value of complementarity without attributing corpus differences to unmeasured traffic properties.

Increasing local fitting data changes the learned local evidence, but the final detector benefits only from changes outside existing peer coverage. On the fixed 58-target cohort, moving from 100 to 400 rows produces small, uncertain end-to-end changes; the 750-row comparison still favors PACE over the comparator. Benefits are weaker in the 47-target and 15-target resource-eligible cohorts, and the high-resource $\alpha = 0.05$ budget change is negative. Consequently, neither larger budgets nor higher standalone local coverage imply monotonic improvement of the final OR rule.

7.4. Practical interpretation

At $\alpha = 0.05$, PACE's primary detection gains accompany lower mean FPR at both budgets. Operational assessment must additionally consider the target distribution: at $\alpha = 0.01$, medians below 0.01 coexist with means above 0.03 and worst-target rates near 0.67. The mean improvement is useful comparative evidence, while the upper tail shows why the nominal parameter cannot serve as a uniform operational guarantee.

Applying the detector requires trusted benign onboarding data for fitting, reference construction, and calibration, plus the preprocessing access in Table 2. References and thresholds remain fixed in the evaluated rule. Independent validation would need to establish when to update them, how to respond to excessive alerts, and how any recalibration changes detection.

Row alerts also require a policy for aggregation into actionable incidents. An anomalous record need not represent an actionable security event [33], and precision depends on attack prevalence [34]. Apruzzese's operational analysis emphasizes the aggregation and triage of reports beyond individual flagged records [35]. The benchmark's separately sampled classes and target-macro rates do not measure incident precision or analyst workload. These operational questions remain distinct from the demonstrated detection contribution.

8. Threats to validity and limitations

Evaluation population and dependence. The 58 eligible targets are not a random deployment sample. Eligibility requires evaluation attacks, sufficient benign calibration data, and enough peers; larger-budget cohorts additionally select for abundant retained training data. Targets can share peer models, captures, collection conditions, and attack-generation procedures. The corpus-stratified bootstrap conditions on the observed corpus groups and does not fully model these dependencies. Three learning seeds assess a fixed design on the same cohort, which was also used during development; they do not sample new corpora. Subgroup and ablation intervals are exploratory, without simultaneous coverage. These boundaries follow established concerns about selection and laboratory evaluation in security machine learning [36]. Benchmark construction can also introduce dataset-specific design artefacts [37]; their impact on PACE must be checked on the retained subsets rather than inferred from corpus names.

Information availability and preprocessing. Dataset annotations identify the benign onboarding roles. Obtaining equally trustworthy observations in deployment remains an assumption, alongside labeled peer attacks and compatible feature schemas. Additional reference and calibration data and full-training-role pooled standardization make N a local fitting budget rather than a total onboarding budget. Gotham's complete-second aggregates can include attack or unlabeled packets, allow later packets to influence earlier rows, and share raw contributors across role boundaries (Section 5). This dependency differs from direct use of attack labels in target fitting. Corpus-specific sampling, deduplication, and source ordering further limit chronological interpretation; separate client and label streams do not establish a network-wide prospective cutoff.

Calibration and distribution change. Calibration controls empirical block rates, without establishing future FPR bounds. The low-alpha allocation exception, reference-tail extrapolation, and final OR without recalibration are part of the implemented rule. Target-level FPR heterogeneity remains substantial under this rule. No online drift detector, adaptive reference update, or recalibration policy was evaluated, and future traffic need not share the calibration distribution.

Comparison and attribution. The primary comparator changes a complete local branch, including detector composition and calibration. Removal ablations also recalibrate the retained experts. They do not isolate every detector's causal contribution or establish optimality of the allocation. Contemporary external competitors, including MCDE, were not executed under the same protocol; the related-work comparison establishes methodological distinctions, not performance superiority. Attack families can occur in peer development data, so the target information restriction is not uniformly an unseen-attack-family test. Transfer is within compatible study strata, not across corpus schemas.

Security and operational deployment. The study assumes trustworthy peers and does not test adversarial-peer behavior, poisoning, evasion, encrypted-feature availability under operational constraints, or attacks on the monitor. Model combination and centralized pooled preprocessing establish no privacy guarantee. End-to-end latency, throughput, memory, energy, communications, incident handling, and online deployment were not measured. The shared benchmark pipeline trains the comparator's autoencoder even though PACE excludes its output; algebraic model sizes cannot substitute for measurements of an eventual deployment implementation.

9. Conclusion

PACE combines separately calibrated supervised peer classifiers with a protected local anchor and a heterogeneous bank of anomaly experts for targets that provide benign observations but no attack labels. Across 58 benchmark targets, replacing a seven-detector mean local branch improves target-macro detection in all four primary configurations. The gains range from approximately 2.32 to 2.83 percentage points, and at nominal $\alpha = 0.05$ they accompany lower mean realized false-alert rates. Rarity-based local evidence supplies the largest measured contribution among the tested expert-family removals, while local-exclusive coverage explains where additional anomaly experts can improve the combined decision.

The results also identify the method's operating boundaries. Gains are concentrated in TON_IoT, become smaller in several other strata, and do not consistently increase with larger local fitting budgets. Realized false-alert rates remain heterogeneous, and the empirical calibration rule does not establish uniform test-time FPR control. The experiments support complementary peer and local detection when the target provides benign data without attack labels. The principal benefit is the additional coverage obtained from separately retained peer decisions and a heterogeneous local branch; operational deployment still requires independent false-alert validation.

Appendix A. Complete Experimental Comparison Inventory

All entries below use the 58-target primary cohort and seeds 6–8. PACE is listed with all 12 executed internal study arms and both recorded baselines. Differences subtract the listed rule's target-macro TPR from PACE's before rounding. Intervals use the same corpus-stratified paired bootstrap as the primary comparison; ablation contrasts are exploratory. The PACE self-comparison has a zero interval by definition. All rates and differences are proportions. Family removals refer to bank experts, not removal of corresponding anchor members.

Table A.1: Definitions of every recorded decision rule.

Rule	Definition
PACE	Separate margin-evidence peer union, independently calibrated six-detector mean anchor, and heterogeneous six-expert bank; $r = a = 0.5$.
Mean local	Primary comparator: identical peer union plus the mean evidence of seven local detectors, including the autoencoder, thresholded at B_L.
No feature	Remove the bank IF, histogram, and Gaussian experts; retain anchor, rarity and response experts, and recalibrate the bank.
No response	Remove the peer-response IF, retaining the anchor and feature/rarity bank; recalibrate the bank.
No rarity	Remove ECOD-sum and pair-surprisal experts; retain anchor, feature and response experts, and recalibrate the bank.
Bank only	Remove the anchor and calibrate the six-expert bank at the entire local allocation B_L; the peer branch is unchanged.
Single union	Jointly calibrate anchor and bank within B_L: initial anchor share a, with $(1 - a)/6$ for each bank expert.
Raw response	Fit the response IF on raw peer margins instead of target-relative evidence; retain the other components and tiers.
Reserve 0.25	Set $r = 0.25$ for the tiered PACE rule, including the implemented anchor-resolution condition; peer allocation is 0.75α.
Peer mean	Mean-fuse peer margin evidence into one calibrated expert at B_P; retain the complete PACE anchor and bank.
Peer max	Maximum-fuse peer margin evidence into one calibrated expert at B_P; retain the complete PACE anchor and bank.
Peers only	Remove the entire local branch and calibrate the separate-peer union at the full α; this is not the extracted B_P peer branch.
Local only	Remove peers and threshold the seven-detector mean local ensemble at the full α; this is not the extracted PACE local branch.
Recorded max	Maximum of probability-tail peer evidence, thresholded once at B_P, plus the seven-detector mean local branch at B_L.
Recorded Bonf.	Separate margin-evidence peers at fixed equal shares of B_P, multiplier one, plus the seven-detector mean local branch at B_L.

Table A.2: Complete nominal comparison results. Δ is PACE minus the listed rule's TPR.

α	N	Rule	TPR	Mean FPR	Δ	95% paired interval
0.01	100	PACE	0.833405	0.033992	0.000000	[0.000000, 0.000000]
0.01	100	Mean local	0.807318	0.030516	0.026086	[0.011323, 0.043346]
0.01	100	No feature	0.829992	0.031778	0.003413	[0.000785, 0.007666]
0.01	100	No response	0.831397	0.033859	0.002008	[0.000277, 0.004518]
0.01	100	No rarity	0.824778	0.033800	0.008626	[0.000602, 0.020504]
0.01	100	Bank only	0.823313	0.033096	0.010092	[-0.017927, 0.052456]
0.01	100	Single union	0.833405	0.033998	0.000000	[0.000000, 0.000000]
0.01	100	Raw response	0.832304	0.033951	0.001101	[-0.001171, 0.003902]
0.01	100	Reserve 0.25	0.814000	0.033265	0.019404	[0.000473, 0.055006]
0.01	100	Peer mean	0.759968	0.026093	0.073437	[0.029755, 0.119872]
0.01	100	Peer max	0.787850	0.025120	0.045555	[0.008586, 0.083883]
0.01	100	Peers only	0.812411	0.030775	0.020993	[-0.011591, 0.056029]
0.01	100	Local only	0.388775	0.019588	0.444630	[0.352850, 0.532565]
0.01	100	Recorded max	0.742730	0.021193	0.090675	[0.045606, 0.136967]
0.01	100	Recorded Bonf.	0.795154	0.029332	0.038251	[0.018699, 0.060976]
0.01	400	PACE	0.838987	0.032369	0.000000	[0.000000, 0.000000]
0.01	400	Mean local	0.810681	0.030781	0.028306	[0.012339, 0.047375]
0.01	400	No feature	0.836974	0.031273	0.002013	[0.000496, 0.003905]
0.01	400	No response	0.836191	0.032074	0.002796	[0.000253, 0.006454]
0.01	400	No rarity	0.827010	0.032295	0.011977	[0.002783, 0.023672]
0.01	400	Bank only	0.824088	0.031255	0.014899	[-0.009737, 0.054637]
0.01	400	Single union	0.838987	0.032371	0.000000	[0.000000, 0.000000]
0.01	400	Raw response	0.837281	0.032211	0.001706	[-0.001722, 0.005636]
0.01	400	Reserve 0.25	0.818840	0.031290	0.020147	[0.001275, 0.055773]
0.01	400	Peer mean	0.783009	0.024896	0.055978	[0.015296, 0.098622]
0.01	400	Peer max	0.802597	0.023855	0.036390	[0.001197, 0.071494]
0.01	400	Peers only	0.812411	0.030775	0.026576	[-0.007605, 0.062007]
0.01	400	Local only	0.475443	0.021635	0.363544	[0.278002, 0.448569]
0.01	400	Recorded max	0.760522	0.022415	0.078465	[0.037711, 0.117340]
0.01	400	Recorded Bonf.	0.798159	0.029579	0.040828	[0.020692, 0.064537]
0.05	100	PACE	0.894767	0.055453	0.000000	[0.000000, 0.000000]
0.05	100	Mean local	0.871348	0.057437	0.023418	[0.005777, 0.043968]
0.05	100	No feature	0.894640	0.052064	0.000127	[-0.002145, 0.002024]
0.05	100	No response	0.891649	0.055375	0.003118	[-0.000083, 0.008302]
0.05	100	No rarity	0.888746	0.055281	0.006020	[0.000008, 0.015617]
0.05	100	Bank only	0.897598	0.060458	-0.002832	[-0.005608, -0.000786]
0.05	100	Single union	0.892561	0.057816	0.002206	[-0.000224, 0.006605]
0.05	100	Raw response	0.891888	0.056676	0.002879	[-0.000354, 0.008044]
0.05	100	Reserve 0.25	0.897066	0.060030	-0.002299	[-0.006719, 0.003283]
0.05	100	Peer mean	0.842308	0.047157	0.052459	[0.021191, 0.091373]
0.05	100	Peer max	0.864283	0.046431	0.030484	[0.014488, 0.048558]
0.05	100	Peers only	0.883945	0.066227	0.010821	[-0.008014, 0.032468]
0.05	100	Local only	0.561134	0.050272	0.333633	[0.249867, 0.419775]
0.05	100	Recorded max	0.815769	0.044953	0.078997	[0.044800, 0.117199]
0.05	100	Recorded Bonf.	0.832897	0.046402	0.061870	[0.034805, 0.093069]
0.05	400	PACE	0.896837	0.054337	0.000000	[0.000000, 0.000000]
0.05	400	Mean local	0.873619	0.064407	0.023219	[0.003116, 0.045968]
0.05	400	No feature	0.896493	0.053259	0.000344	[-0.000063, 0.000897]

Table A.2 continued

α	N	Rule	TPR	Mean FPR	Δ	95% paired interval
0.05	400	No response	0.895568	0.054558	0.001269	[0.000021, 0.003571]
0.05	400	No rarity	0.886434	0.053995	0.010403	[0.000237, 0.023050]
0.05	400	Bank only	0.898372	0.064650	-0.001535	[-0.003242, -0.000329]
0.05	400	Single union	0.897186	0.065082	-0.000349	[-0.000868, 0.000045]
0.05	400	Raw response	0.895565	0.058334	0.001272	[0.000021, 0.003573]
0.05	400	Reserve 0.25	0.901835	0.060317	-0.004998	[-0.008622, -0.001879]
0.05	400	Peer mean	0.852114	0.044361	0.044723	[0.017602, 0.080736]
0.05	400	Peer max	0.865193	0.043049	0.031644	[0.016224, 0.049405]
0.05	400	Peers only	0.883945	0.066227	0.012892	[-0.007048, 0.036236]
0.05	400	Local only	0.628398	0.055778	0.268439	[0.189487, 0.351664]
0.05	400	Recorded max	0.822336	0.052848	0.074501	[0.040334, 0.111457]
0.05	400	Recorded Bonf.	0.839911	0.051129	0.056926	[0.028144, 0.090626]

Appendix B. Dataset inputs and provenance

Table B.1 identifies the evaluated inputs from client-cache manifests and the adapters in `src/pace/datasets/adapters/`. A local directory name such as `IoT-23-v2` is not evidence of a publisher version 2. The input manifest in the manuscript supplement lists exact source paths, feature-schema identifiers, and independently recomputed SHA-256 digests of the seven cache manifests. These identify inspected metadata, not an independent checksum validation of every raw archive. Campaign content digests and the recorded input/preprocessing digests identify frozen evaluation provenance; they do not establish current raw-source byte identity.

Table B.1: Dataset releases, evaluated subsets, and source access. Links refer to authoritative source pages or deposits; no third-party raw data are redistributed with the manuscript.

Dataset	Evaluated input and adapter	Source and conditions
IoT-23	`iot_23_datasets_small.tar.gz`: labeled Zeek flows, not the full packet-capture collection. Dataset record v1.0.0; adapter `iot23.py`.	Stratosphere source page; Zenodo record, CC BY 4.0.
TON_IoT	`Processed_datasets/Processed_Network_dataset`; network CSVs, not telemetry or OS traces. No numbered release recorded in the cache; adapter `ton_iot.py`.	UNSW source page: academic research use granted; commercial use requires the owner's permission; source citation conditions apply.
Gotham	2025 Zenodo deposit; eight device CSV inputs under `extracted/processed/`, seven reported targets plus one peer-only client. Adapter `gotham.py`; full-second features described in Section 5.	Zenodo dataset, CC BY 4.0; the 2026 descriptor cites this deposit [29].

Dataset	Evaluated input and adapter (continued)	Source and conditions
CTU-13	`CTU-13-Dataset.tar.bz2`; 12 eligible scenario targets from the original 13-scenario collection. Adapter `ctu13.py` uses labeled Argus flows and discards Background. No separate archive revision is recorded.	Stratosphere CTU-13 page; the source FAQ permits use with attribution; separate raw redistribution terms are not verified here.
N-BaIoT	Original 2018 nine-device release; extracted benign and Mirai/BASHLITE feature CSVs. Adapter `nbaiot.py`, including within-file exact-row deduplication.	UCI dataset record [31], CC BY 4.0; distinct from the original research paper [15].
UNSW IoT attack flows	SOSR 2019 release: ten device `*_flowstats.csv` files and associated annotation inputs under `flowdata/flowdata/`, split into G1/G2 compatible feature schemas. Adapter `unsw.py`.	UNSW attack-traces page: minute-level MUD flow counters and annotations; no explicit reuse licence verified on that page.

The manifests do not record retrieval dates or numbered revisions for the TON_IoT and UNSW inputs, nor independently verified raw-archive checksums. **Author confirmation required:** any additional archive-version or retrieval records and applicable redistribution permissions not established by the cited source terms. These gaps do not change the identity of the frozen campaign artifacts.

Data and code availability

The implementation, experimental configurations, and frozen evaluation artifacts are maintained in the PACE project repository. The permanent public repository address and immutable revision identifier will be supplied before submission. Third-party datasets remain subject to their original access and redistribution conditions. Dataset inputs and access conditions are identified in Appendix B. **Author-supplied release link and immutable revision: pending.**

Funding and competing interests

Funding sources and competing-interest declarations require confirmation by the authors. No absence of funding or competing interests is inferred.

Authorship and acknowledgements

Author confirmation required: individual contribution roles, any acknowledgements, and approval of the final manuscript and declarations.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex was used for repository inspection, literature assistance, manuscript drafting, and technical verification. This draft requires human scientific review and approval. The authors must confirm the extent of their review and responsibility for the final content before submission; this draft does not assert that such approval has already occurred.

References

[1] A. Zimek, R. J. Campello, J. Sander, Ensembles for Unsupervised Outlier Detection: Challenges and Research Questions [Position Paper], ACM SIGKDD Explorations Newsletter 15 (1) (2014) 11–22. doi:10.1145/2594473.2594476.

[2] J. Yue, Y. Lu, Y. Xie, Detecting unseen IoT attacks with calibrated dual evidence under low false-positive budget, Entropy 28 (9) (2026) 1026. doi:10.3390/e28091026.

[3] Y. Mirsky, T. Doitshman, Y. Elovici, A. Shabtai, Kitsune: An Ensemble of Autoencoders for Online Network Intrusion Detection, in: Proceedings 2018 Network and Distributed System Security Symposium, 2018. doi:10.14722/ndss.2018.23204.

[4] T. D. Nguyen, S. Marchal, M. Miettinen, H. Fereidooni, N. Asokan, A.-R. Sadeghi, DÏoT: A Federated Self-learning Anomaly Detection System for IoT, in: 2019 IEEE 39th International Conference on Distributed Computing Systems (ICDCS), 2019, pp. 756–767. doi:10.1109/icdcs.2019.00080.

[5] J. Wu, H. Dai, K. B. Kent, J. Yen, C. Xu, Y. Wang, Open Set Dandelion Network for IoT Intrusion Detection, ACM Transactions on Internet Technology 24 (1) (2024) 1–26. doi:10.1145/3639822.

[6] A. N. Angelopoulos, S. Bates, Conformal prediction: A gentle introduction, Foundations and Trends in Machine Learning 16 (4) (2023) 494–591. doi:10.1561/2200000101.

[7] S. Bates, E. Candès, L. Lei, Y. Romano, M. Sesia, Testing for outliers with conformal p-values, The Annals of Statistics 51 (1) (2023) 149–178. doi:10.1214/22-AOS2244.

[8] V. Vovk, R. Wang, Combining p-values via averaging, Biometrika 107 (4) (2020) 791–808. doi:10.1093/biomet/asaa027.

[9] R. F. Barber, E. J. Candès, A. Ramdas, R. J. Tibshirani, Conformal prediction beyond exchangeability, The Annals of Statistics 51 (2) (2023) 816–845. doi:10.1214/23-AOS2276.

[10] T. Xiang, A. Yang, G. So, Target-adaptive CAN intrusion detection across heterogeneous vehicle datasets via robust residual modeling and empirical block-rank calibration, Preprints.org, version 1, posted 18 September 2026; not peer reviewed (2026). doi:10.20944/preprints202609.1557.v1.

[11] R. A. Bridges, J. D. Jamieson, J. W. Reed, Setting the threshold for high throughput detectors: A mathematical approach for ensembles of dynamic, heterogeneous, probabilistic anomaly detectors, in: 2017 IEEE International Conference on Big Data (Big Data), 2017, pp. 1071–1078. doi:10.1109/BigData.2017.8258031.

[12] F. T. Liu, K. M. Ting, Z.-H. Zhou, Isolation forest, in: 2008 Eighth IEEE International Conference on Data Mining, IEEE, 2008, pp. 413–422. doi:10.1109/ICDM.2008.17.

[13] M. M. Breunig, H.-P. Kriegel, R. T. Ng, J. Sander, LOF: Identifying density-based local outliers, in: Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data, ACM, 2000, pp. 93–104. doi:10.1145/342009.335388.

[14] G. Pang, C. Shen, L. Cao, A. van den Hengel, Deep learning for anomaly detection: A review, ACM Computing Surveys 54 (2) (2021) 38:1–38:38. doi:10.1145/3439950.

[15] Y. Meidan, M. Bohadana, Y. Mathov, Y. Mirsky, A. Shabtai, D. Breitenbacher, Y. Elovici, N-BaIoT—Network-Based Detection of IoT Botnet Attacks Using Deep Autoencoders, IEEE Pervasive Computing 17 (3) (2018) 12–22. doi:10.1109/mprv.2018.03367731.

[16] V. T. Nguyen, R. Beuran, FedMSE: Semi-supervised federated learning approach for IoT network intrusion detection, Computers & Security 151 (2025) 104337. doi:10.1016/j.cose.2025.104337.

[17] V. Rey, P. M. Sánchez Sánchez, A. Huertas Celdrán, G. Bovet, Federated learning for malware detection in IoT devices, Computer Networks 204 (2022) 108693. doi:10.1016/j.comnet.2021.108693.

[18] C. S. Abhijit, Y. A. Jerusha, S. P. S. Ibrahim, V. Varadharajan, A personalized federated hypernetworks based aggregation approach for intrusion detection systems, Scientific Reports 15 (1) (2025) 33974. doi:10.1038/s41598-025-11659-7.

[19] S. Layeghy, M. Baktashmotlagh, M. Portmann, DI-NIDS: Domain invariant network intrusion detection system, Knowledge-Based Systems 273 (2023) 110626. doi:10.1016/j.knosys.2023.110626.

[20] J. Talpini, N. Civiero, F. Sartori, M. Savi, A Federated Approach to Enhance Calibration of Distributed ML-Based Intrusion Detection Systems, in: Proceedings of the 17th International Conference on Agents and Artificial Intelligence, 2025, pp. 840–848. doi:10.5220/0013376600003890.

[21] H. Wang, Z. Li, L. Feng, W. Zhang, ViM: Out-Of-Distribution with Virtual-logit Matching, in: 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 4911–4920. doi:10.1109/cvpr52688.2022.00487.

[22] H. Jin, Cross-protocol domain gap in internet of things intrusion and anomaly detection: An empirical internet protocol-to-bluetooth low energy study of domain-adversarial training, Sensors 26 (4) (2026) 1184. doi:10.3390/s26041184.

[23] Y. Ha, K. Kim, Operationally constrained zero-day intrusion detection with Target-FPR calibration and similarity graph construction, Applied Sciences 16 (5) (2026) 2284. doi:10.3390/app16052284.

[24] M. Nouman, M. Hassanu, R. Ujjan, Calibration Granularity, Not Contamination: Diagnosing a TCN Anomaly Detector's False Positive Advantage in Cross-Dataset IoT Traffic, Future Internet 18 (9) (2026) 447. doi:10.3390/fi18090447.

[25] Z. Li, Y. Zhao, X. Hu, N. Botta, C. Ionescu, G. H. Chen, ECOD: Unsupervised Outlier Detection Using Empirical Cumulative Distribution Functions, IEEE Transactions on Knowledge and Data Engineering 35 (12) (2023) 12181–12193. doi:10.1109/tkde.2022.3159580.

[26] S. Garcia, A. Parmisano, M. J. Erquiaga, IoT-23: A labeled dataset with malicious and benign IoT network traffic, Dataset, Zenodo (2020). doi:10.5281/zenodo.4743746.

[27] N. Moustafa, A new distributed architecture for evaluating AI-based security systems at the edge: Network TON_IoT datasets, Sustainable Cities and Society 72 (2021) 102994. doi:10.1016/j.scs.2021.102994.

[28] O. Belarbi, T. Spyridopoulos, E. Anthi, O. Rana, P. Carnelli, A. Khan, A device-level IoT network traffic dataset with distributed capture and Non-IID characteristics, Dataset, version 1, Zenodo (2025). doi:10.5281/zenodo.14502760.

[29] O. Belarbi, T. Spyridopoulos, E. Anthi, O. Rana, P. Carnelli, A. Khan, A Device-Level IoT Network Traffic Dataset with Distributed Capture and Non-IID Characteristics, Data 11 (8) (2026) 207. doi:10.3390/data11080207.

[30] S. García, M. Grill, J. Stiborek, A. Zunino, An empirical comparison of botnet detection methods, Computers & Security 45 (2014) 100–123. doi:10.1016/j.cose.2014.05.011.

[31] Y. Meidan, M. Bohadana, Y. Mathov, Y. Mirsky, D. Breitenbacher, A. Shabtai, detection_of_IoT_botnet_attacks_N_BaIoT, Dataset, UCI Machine Learning Repository (2018). doi:10.24432/C5RC8J.

[32] A. Hamza, H. H. Gharakheili, T. A. Benson, V. Sivaraman, Detecting volumetric attacks on IoT devices via SDN-based monitoring of MUD activity, in: Proceedings of the 2019 ACM Symposium on SDN Research, 2019, pp. 36–48. doi:10.1145/3314148.3314352.

[33] R. Sommer, V. Paxson, Outside the closed world: On using machine learning for network intrusion detection, in: 2010 IEEE Symposium on Security and Privacy, 2010, pp. 305–316. doi:10.1109/SP.2010.25.

[34] T. Saito, M. Rehmsmeier, The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets, PLOS ONE 10 (3) (2015) e0118432. doi:10.1371/journal.pone.0118432.

[35] G. Apruzzese, SoK: Reshaping Research on Network Intrusion Detection Systems, in: Proceedings of the ACM Asia Conference on Computer and Communications Security, 2026, pp. 1028–1042. doi:10.1145/3779208.3807488.

[36] D. Arp, E. Quiring, F. Pendlebury, A. Warnecke, F. Pierazzi, C. Wressnegger, L. Cavallaro, K. Rieck, Pitfalls in machine learning for computer security, Communications of the ACM 67 (11) (2024) 104–112. doi:10.1145/3643456.

[37] R. Flood, G. Engelen, D. Aspinall, L. Desmet, Bad Design Smells in Benchmark NIDS Datasets, in: 2024 IEEE 9th European Symposium on Security and Privacy (EuroS&P), 2024, pp. 658–675. doi:10.1109/EuroSP60621.2024.00042.